

Model-free: Monte Carlo estimation

MC Basic

$$Q_{\pi_k}(s, a) = \sum_r \underbrace{p(r|s, a)}_{\text{model-based}} r + \gamma \sum_{s'} \underbrace{p(s'|s, a)}_{\text{model-based}} V_{\pi_k}(s')$$

↓ model-free

$$Q_{\pi_k}(s, a) = E[G_t | S_t = s, A_t = a]$$

Algo (MC Basic)

Init: π_0

Iteration:

① PE: for $V(s, a)$, get $Q_{\pi_k}(s, a)$ Suppose a set of episodes $\{g^{(j)}(s, a)\}$ experience

then

$$Q_{\pi_k}(s, a) \approx \frac{1}{N} \sum_{j=1}^N g^{(j)}(s, a)$$

② PI: let $a_k^*(s) = \arg\max_a Q_{\pi_k}(s, a)$, $s \in S$.

$$\Rightarrow \pi_{k+1}(a_k^* | s) = 1$$

set episode length

MC Exploring Starts

+ ① $s_1 \xrightarrow{a_1} s_2 \xrightarrow{a_2} s_3 \xrightarrow{a_3} \dots$ { first-visit
every-visit

② 1 episode $\rightarrow$ 1 updateMC ϵ -greedy

ϵ -greedy policy: $\pi(a|s) = \begin{cases} 1 - \frac{\epsilon}{|A(s)|} (|A(s)| - 1), & a = a^* \text{ greedy action.} \\ \frac{\epsilon}{|A(s)|}, & \text{else.} \end{cases}$

$\epsilon \downarrow$ greedy $\uparrow$
 $\epsilon \uparrow$ exploration $\uparrow$

1 episode, long steps